

ECE 269: Linear Algebra and Applications

Prof. Piya Pal

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Logistics

Logistics: Lectures and Discussion

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- **Interactions during lecture:** Questions during the lecture are highly encouraged. Due to time limitations, I will restrict them to prescribed intervals towards the middle and end of the lecture. Post-lecture questions are very much welcome during office hours, and/or on Canvas (by creating "Discussion Threads").

Course Components and Syllabus

Instructor and TAs

Instructor: Prof. Piya Pal

Email: pipal@ucsd.edu

Office Hours: Tuesdays and Thursdays: 1:00 - 2:00 pm PDT, FAH 2002 and [Zoom](#)

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Teaching Assistants:

- ① Yinyan Bu (Head TA), y1bu@ucsd.edu
Office Hours: Tuesdays 3:00 - 4:00 pm PDT, Jacobs Hall 5101E and Zoom
- ② Mohamed Hamdy, mohamdy@ucsd.edu
Office Hours: Mondays and Thursdays 2:00 - 3:30 am PDT, Jacobs Hall 5101E and Zoom
- ③ Minhong Zhou, miz152@ucsd.edu
Office Hours: Fridays 2:00 - 4:00 pm PDT, Jacobs Hall 5101D and Zoom

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Website: Managed Via [Canvas](#) integrated with [Zoom](#) and [Gradescope](#)

Homework, Project, Exam and Grading

Grading: Attendance: 5%, Quizzes 15%, Mini Projects 15%, MidTerm 25%,
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Quizzes: We will share around 4 homework sets and their solutions. We will conduct 2-3 in-class (during lectures or discussions) quizzes to test your understanding of the homeworks. Quizzes will be closed-book, no electronic devices allowed.

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Homeworks and Quizzes

- We will not grade the homeworks. The problems and solutions of each homework will be posted (one week apart) for self study.
- We highly encourage you to try to solve the questions on your own, without looking at the solutions. You should bring questions about the homeworks to our office hours to get any potential misconceptions clarified.
- The quizzes will be directly based on the homework problems and test your understanding.

Exam and Grading Policies

- All exams and quizzes are in-person, closed book with no electronic devices allowed.
- There will be no make-up exams. It is your responsibility to ensure that you do not have a schedule conflict involving the examinations (you should not enroll in this class if you cannot sit for the examinations at their scheduled times).
- Depending on how most students perform in the quizzes, we may drop the quiz with the lower score.
- We may adapt our grading policies in order to help students. Prof. Pal reserves the right to do so.

Policy on the use of AI

- AI has emerged a powerful educational tool. But like any tool, it can be used in “smart” and “not so smart” ways. You can use it to refine your understanding of a concept, or the solution of a problem. But...

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- Exercise your natural intelligence, and always remember the “No Free Lunch” theorem.
- **Use of AI is prohibited for mini projects.** If we suspect the use of AI in mini projects, we will report you to the office of academic integrity, or award you a score of zero in the project (or both).

References

Following is a list of suggested references (basic and advanced). [Prof. Pal will post .pdf copies of lecture notes from every lecture](#), as well as supplementary research papers and articles as required. You are expected to pay attention during lectures and encouraged to take notes.

- C. D. Meyer, Matrix Analysis and Applied Linear Algebra, SIAM 2000 [Basic/Advanced, [Recommended](#)].

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- G. Strang, Linear Algebra and Learning from Data, Wellesley-Cambridge Press, 2019 [Basic].
- S. Lang, Linear Algebra, Springer, 3rd Ed., 2004 [Basic].
- Peter D. Lax, Linear Algebra and its Applications, Wiley-Interscience, 2007 [Advanced].
- Horn and Johnson, Matrix Analysis, Cambridge University Press, 2012 [Advanced].

- **Linear Algebra (MIT Open courseware):** A classic course by Prof. Gilbert Strang.
<https://ocw.mit.edu/courses/mathematics/18-06-linear-algebra-spring-2010>
Contains video lectures. May serve as a helpful refresher for some of the topics.
- **EE263 (Stanford):** A course on Linear Dynamical Systems.
<http://ee263.stanford.edu>
- **The Matrix Cookbook:** Handy reference for matrix calculus.
<http://www2.imm.dtu.dk/pubdb/pubs/3274-full.html>
- You can “Google” any topic covered in lecture or use Wikipedia, but you should be careful in accepting or rejecting the result of your findings.

Advice and Tips for Success: I

- **Start Early:** Start homeworks early! Do not look at solutions. “Sleep over” the problems. Bring your questions to office hours (Prof. Pal and TAs). Start new “Discussion” threads on Canvas, which TAs will frequently monitor.

Advice and Tips for Success: I

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- **(Most Important) Invest Time in Problem Solving:** Let the homework problem sink in. Invest sufficient time so that you can finally get to the “heart of the problem”, and reflect on it. Resist the temptation to “look” for a ready-made solution. Invest that time/effort towards trying to solve it on your own before asking anyone (including Google). The former may provide a short-term gain, but the latter will lead to deeper understanding.

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- **Take notes:** I encourage students to take notes whenever possible (that is how I learned, and still learn new topics). Studies have shown that one retains information better in this way. You can also refine your notes when you revisit them after lectures, which will really help in solidifying your understanding.

Advice and Tips for Success: II

- **Make Mistakes (and be Vigilant!):** Do not be afraid of making mistakes. Think of them as an essential part of learning, one that leads to lasting wisdom! You learn way better by first making mistakes, and then trying to find a way out. That way, you know “what does not work” (in addition to knowing what finally worked). Make mistakes, and learn from them while trying to solve homework problems. Also remember: what seems like a “small” mistake, may not be small after all. It can hide deeper conceptual errors.

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- **Know where to get help from in times of need:** Know your resources. This includes Prof. Pal and the TAs, Discussion on Canvas, and your friends. Follow my previous advice on “Invest Time in Problem Solving”, so that you really benefit from these resources. Discussing the nuances of a problem with friends, and trying to teach each other, are good ways to learn. You should uphold academic integrity and obey homework policies during such interactions.

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- **Less is More:** Try to do fewer things as accurately as you can (instead of “solving” many problems halfway). This approach will make you a perfectionist, and help you tremendously during exams.

Course Objectives and Topics

Course Objective

- Representation, manipulation of signals/data, and inferring from them, are fundamental goals in problems arising in signal processing, communication, statistics, machine learning, and data sciences. Linear algebra constitutes the main mathematical workforce behind majority of these methods. Broadly, linear algebra is the study of mathematical principles guiding linear operations with particular reference to the geometry of vector spaces over which such operations are defined. When such operations are performed in finite dimensional vector spaces, they can be defined in terms of matrix operations. In this course, we will **build the mathematical foundations of linear algebraic techniques which will justify their use in diverse applications.**

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- Through the lens of linear algebra, you will hopefully learn to appreciate the **power, versatility (and beauty) of mathematical abstractions**, which once mastered, can be adapted (relatively easily) to solve specific real-world scientific/engineering problems arising in apparently disconnected application domains.

Topics: Theory

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- Additional Topics (Time permitting): Kernel Methods, Non-negative Matrices.

Topics: Applications

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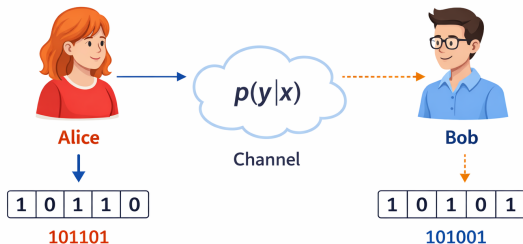
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- **Communication:** MIMO Communication, Coding theory (Linear Algebra over Finite Fields), Physical Layer Optimization.
- **Complex Networks/Graphs:** Spectral graph theory, Graph Embeddings, Page Rank algorithm.

Applications: Wireless Communication



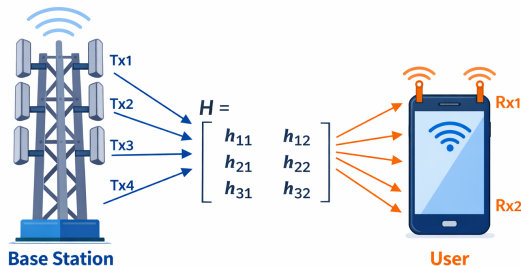
- Reliable communication over an unreliable noisy (wireless) channel.
- **Channel coding for error correction:** builds on linear algebra.

Applications: Image/Video Compression



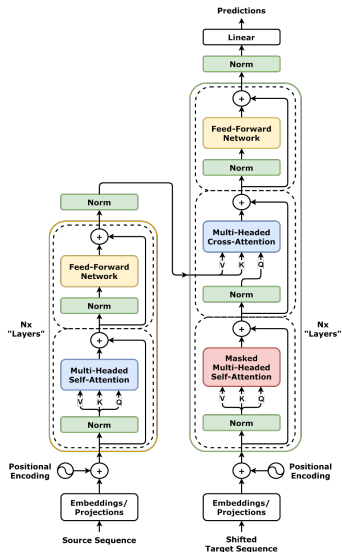
- Image compression involves representation of manipulation of images in new domains.
- The transformation happens using linear algebraic principles, combined with source coding.

Applications: Wireless Communication



- Multiple-input-multiple-output channel is a matrix.
- **Channel estimation, equalization, precoding, beamforming, signaling strategies, symbol detection:** Extensive use of linear algebra.

Applications: Machine Learning



Attention is all you need!

- Attention mechanism is a simple linear algebraic principle, which you will learn later in the course (but with profound implications).
- Transformers heavily use linear algebraic principles and matrix computations.

Academic Integrity and Copyright

Academic Integrity

- In order to help promote academic integrity, collaboration policies on every homework, project and final exam will be clearly stated. Any violation or suspected academic dishonesty (such as cheating during exam, copying another student's solution to homework problems etc) will be duly reported to the Office of Academic Integrity.
- Serious instances of academic dishonesty generally results in a grade of F/academic probation.
- Under certain circumstances, it can also lead to a permanent record of offense on the student's academic record, which may be visible to future employers.

Academic Integrity

“Academic Integrity is expected of everyone at UC San Diego. This means that you must be honest, fair, responsible, respectful, and trustworthy in all of your actions. Lying, cheating or any other forms of dishonesty will not be tolerated because they undermine learning and the University’s ability to certify students’ knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying or dishonesty will be reported to the Academic Integrity Office and will result sanctions. Sanctions can include an F in this class and suspension or dismissal from the University. So, think carefully before you act by asking yourself: a) is what I’m about to do or submit for credit an honest, fair, respectful, responsible trustworthy representation of my knowledge and abilities at this time and, b) would my instructor approve of my action? You are ultimately the only person responsible for your behavior. So, if you are unsure, don’t ask a friend—ask your instructor, instructional assistant, or the Academic Integrity Office. You can learn more about academic integrity at academicintegrity.ucsd.edu” (Source: Academic Integrity Office, 2018)

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- If you do so, you may be subject to student conduct proceedings under the UC San Diego Student Code of Conduct.
- Similarly, you own the copyright in your original papers and exam essays. If I am interested in posting your answers or papers on the course web site, I will ask for your written permission.

Questions?